

NKB PHARMACEUTICALS & COSMETICS MANUFACTURING

B2B Order Lifecycle, Cleanroom Batching & Universal Traceability Concept Map

High-fidelity blueprint illustrating end-to-end order orchestration from client purchase order creation, automated tolerance validation ($\pm 10\%$), cleanroom compounding & bottling with 106-member employee roster assignment, overrun buffer logic, digital delivery acceptance, and sales invoice reconciliation.

Standard: ISO 22716 GMP Cosmetics
Target Tolerance: $\pm 10.0\%$ Allowable
System Version: v2.4 (Active Live)
Date Generated: September 12, 2026

1 Phase I: Order Ingestion, Production Allocation & Staff Assignment

STEP 01 • CLIENT PO

Purchase Order

Client submits target quantities.
Contract pricing rules (e.g. ₱180.00/pc) and $\pm 10\%$ tolerance bounds are locked.

Doc ID:	PO-2026-000001
Policy:	Actual Delivery
Target:	500 pcs [450-550]

STEP 02 • PURCHASING REQUISITION

Raw Materials & Requisition

Inventory evaluates chemical actives & packaging stock. Shortages trigger requisitions for Purchasing Department to order and deliver.

Req Dept:	Purchasing
Materials:	Kojic Acid, 50g Tubes
Status:	DELIVERED -> SUFFICIENT

STEP 03 • ADMIN JOB ORDER

Job Order (JO)

Approved PO with confirmed raw materials routes to plant scheduling. Formula specifications locked ('FORM-SGC-V1') and cleanroom team assigned.

Doc ID:	JO-2026-000001
Team:	Formulation Beta
Action:	 View PO Button

STEP 04 • STAFF & LINE ASSIGN

Roster Allocation

Cleanroom stations populated from 106-staff roster: Compounding Operator, Bottling Lead, QC Officer & Line designation.

Compounder:	Catindig, Renar A.
Bottling:	Alonzo, Merry Jean
Line:	Cleanroom Line 1

2 Phase II: Quality Clearance, Dispatch, Digital DR & Settlement

STEP 05 • YIELD & OVERRUN

Batching & Yield

Actual yield logged. System computes variance vs $\pm 10\%$ threshold. Automated overrun approval logic engages.

Batch ID:	BAT-2026-000001
Actual Yield:	550 pcs (+10.0%)
Status:	TOLERANCE_PASS

STEP 06 • QC INSPECTION

QC Inspection Station

QC Inspector performs viscosity, pH stability (5.6), SPF testing, and sterility tests. Batch clearance released with Certificate of Analysis (CoA).

Inspector:	Fabio, Marilou (QC)
Result:	QC_PASSED / Micro Cleared
Artifact:	CoA Issued & Signed

STEP 07 • DELIVERY (DR)

Warehouse Dispatch

Delivery Receipt created. Packaged with lot numbers and expiry dates. Driver dispatched under vehicle logs.

Doc ID:	DR-2026-000001
Driver:	Danilo Gomez (NKB-8899)
Delivered:	550 pcs

STEP 08 • ACCEPTANCE & INVOICE

Sign-off & Settlement

Client signs digital DR acceptance. Sales Invoice generated for accepted count (550 pcs x ₱180 = ₱99,000) and payment recorded.

Invoice ID:	SI-2026-000001
Total / Paid:	₱99,000 / ₱50,000
Payment Ref:	PAY-2026-000001 (BDO)

Universal "View PO / View Order" Traceability Engine

In cosmetic manufacturing, complete auditability is legally mandated under FDA and GMP standards. NKB Client V2 implements **Universal Order Traceability**: Every operational node—Job Orders, Production Batches, Delivery Receipts, Client Buffer Stock, and Sales Invoices—features a persistent " **View PO**" button.

With 1 click, operators and auditors can inspect the root contract terms, agreed unit pricing (with decimal precision), target vs allowable bounds, and complete production lineage. From this modal, a dedicated " **Print Order Info**" action generates instant, formatted physical and PDF order records.

TRACEABILITY LINEAGE MODAL

LIVE AUDIT

- Root PO: **PO-2026-000001** (SKEENCARE)
- Unit Price: **₱180.00** | Policy: **ACTUAL_DELIVERY**
- Job Order: **JO-2026-000001** [Formulation Beta]
- Batch Run: **BAT-2026-000001** [550 pcs / +10.0%]
- Compounder: **Catindig, Renar A.**
- Delivery DR: **DR-2026-000001** [Danilo Gomez / NKB-8899]
- Accepted By: **Maria Santos** [Digital Signature]
- Sales Invoice: **SI-2026-000001** [₱99,000.00 Total]

Rule 1: Standard Tolerance Window ($\pm 10.0\%$)

Formulation vessels naturally fluctuate based on evaporation and clean-out. Target quantities automatically generate an acceptable manufacturing envelope:

$\text{Min} = \text{Target} \times 0.90$ | $\text{Max} = \text{Target} \times 1.10$. Batches landing inside this envelope are cleared for standard dispatch without administrative bottleneck.

Rule 2: Actual Delivery vs Fixed Billing

Actual Delivery: Client accepts and pays for the exact produced quantity (e.g. 550 pcs @ ₱180 = ₱99,000).

Fixed PO: Client pays only for the base target (500 pcs @ ₱180 = ₱90,000); surplus overrun (50 pcs) is securely diverted to the client's dedicated **Client Buffer Stock** vault.

Rule 3: Exception & Overrun Threshold ($> +10\%$)

When yield exceeds $+10\%$, the batch automatically transitions to `EXCEPTION_REQUIRES_APPROVAL`. The system halts automated DR dispatch until an Admin and the Client provide cryptographic overrun sign-off or allocate the surplus to future buffer stock.

Enterprise Multi-Agent Collaboration Suite

To ensure zero operational lag between plant departments and enterprise clients, NKB Client V2 features two autonomous floating agents:

- **Action Notification Agent** (🔔): Dynamically calculates role-tailored work queues (Purchasing requisitions, Accounting confirmations, Inventory checks, QC batch clearances, Warehouse DR dispatches). Clicking any pending task navigates directly to the exact record.
- **Enterprise Chat System** (💬): Encrypted cross-departmental messenger enabling instant internal communication across all roles, with strict security boundaries restricting external clients to the Accounting department and a pinned direct channel to Super Admin & IT Support.

AUTONOMOUS AGENT ACTIVE CHANNELS

LIVE POLLING

- 🔔 **Action Agent:** Real-time task queue per role
- 💬 **Enterprise Messenger:** Internal cross-department
- 🛡️ **Client Boundary:** Accounting & Support ONLY
- 👑 **Executive Dashboard:** Real-time CEO oversight
- 🧪 **QC Station:** Batch sampling & CoA verification
- 🛒 **Purchasing:** Auto PO material unblocking

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Cleanroom Roles & System Roster Integration (106 Members)

Role / Station	Assigned Personnel	Operational Responsibilities	System Verification Artifact
Chief Executive Officer (CEO)	Executive Oversight (CEO)	Omniscient visibility across all manufacturing facilities, purchase orders, compounding lines, delivery receipts, client stock buffer, and financial ledgers.	Executive Dashboard & Global Audit Lineage
Purchasing Department (Procurement)	Purchasing Officer	Evaluates inventory raw materials requisitions, sources chemical actives & custom bottles, tracks supplier lead times, and receives deliveries into warehouse stock.	Manages supply_requests & unblocks PO materials to SUFFICIENT
Compounding Operator	Catindig, Renar A.	Bulk vessel weighing, heating, homogenization, and ingredient addition according to master batch records.	Logs raw material usage, vessel IDs, and initial formulation yield.
Bottling & Packaging Lead	Alonzo, Merry Jean I.	Fills primary containers (tubes/jars), performs seal torquing, batch coding, and master carton packaging.	Logs final piece count yield (550 pcs) and packaging station checks.
Quality Control (QC) Inspector	Fabio, Marilou (QC)	Physicochemical testing (pH, viscosity, specific gravity), microbial clearance, and SPF retention verification. Authorizes batch clearance.	Authorizes QC_PASSED status and issues Certificate of Analysis (CoA).
Warehouse & Logistics	Danilo Gomez / wh_admin	Pick & pack, palletizing, loading vehicle NKB-8899, DR generation, and client site delivery dispatch.	Dispatches DR-2026-000001 with driver and plate number.
Client Purchasing Officer	Maria Santos (SKEENCARE)	Inspects physical shipment at client receiving dock, verifies seals, and executes digital acceptance.	Submits digital signature and creates dr_acceptances record.

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Role / Station	Assigned Personnel	Operational Responsibilities	System Verification Artifact
Finance & Accounting	Accounting Specialist	Calculates line totals based on billing policy (Actual vs Fixed), generates SI, and records payments.	Issues SI-2026-000001 and reconciliation PAY-2026-000001.